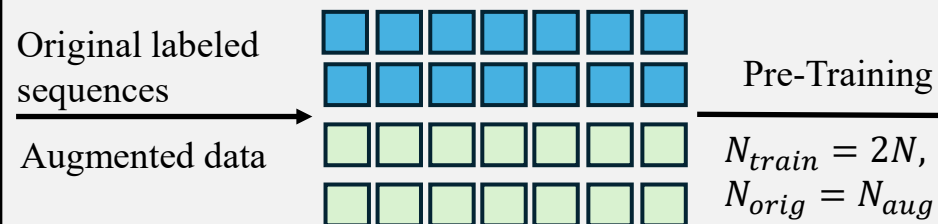


E. Two stage Supervised training

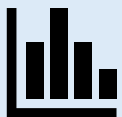
Stage1: pre-training



Stage 2: fine-tuning



F. Held-out Evaluation



- Evaluate on original held-out examples only.
- Compare validation AUC for each augmentation arm
- Summarize paired Δ AUC across tasks

1D Residual CNN

Input: One-hot protein sequence
 $20 \times L$

Stem

Conv1D($20 \rightarrow 64, k=9$) + BN + ReLU + MaxPool

Residual stage 1

2 residual blocks, 64 channels

Residual stage 2

projection block + residual block, 128 channels

Residual stage 3

Projection block + residual block, 256 channels

Adaptive GAP $\rightarrow$ Linear
binary prediction